

4. The North American black bear (*Ursus americanus*) is widespread through the USA and Canada. In British Columbia a form of the North American black bear is found which is white or cream in colour. This form of the North American black bear is known as the Kermode or spirit bear. This is shown in **Image 4.1**. [This is not the same species as the polar bear (*Ursus maritimus*).]

Image 4.1



Kermode or spirit bear

The white or cream colour of the Kermode bear is due to a recessive allele.

- (a) (i) Two heterozygous North American black bears were mated. Using the letters **B** and **b**, complete the Punnett square below to show this cross. [2]

F1	Gametes		
			
			
			

- (ii) In the Punnett square **draw a circle** around the genotype of a Kermode bear offspring. [1]



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(iii) North American black bears reproduce every two years and produce 2–3 offspring at a time. Explain why, if they only produce 2–3 offspring at a time, the Punnett square contains boxes for four offspring. [2]

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(b) Explain why scientists use the scientific names *Ursus americanus* and *Ursus maritimus* when describing bears rather than the common names North American black bear and polar bear respectively. [1]

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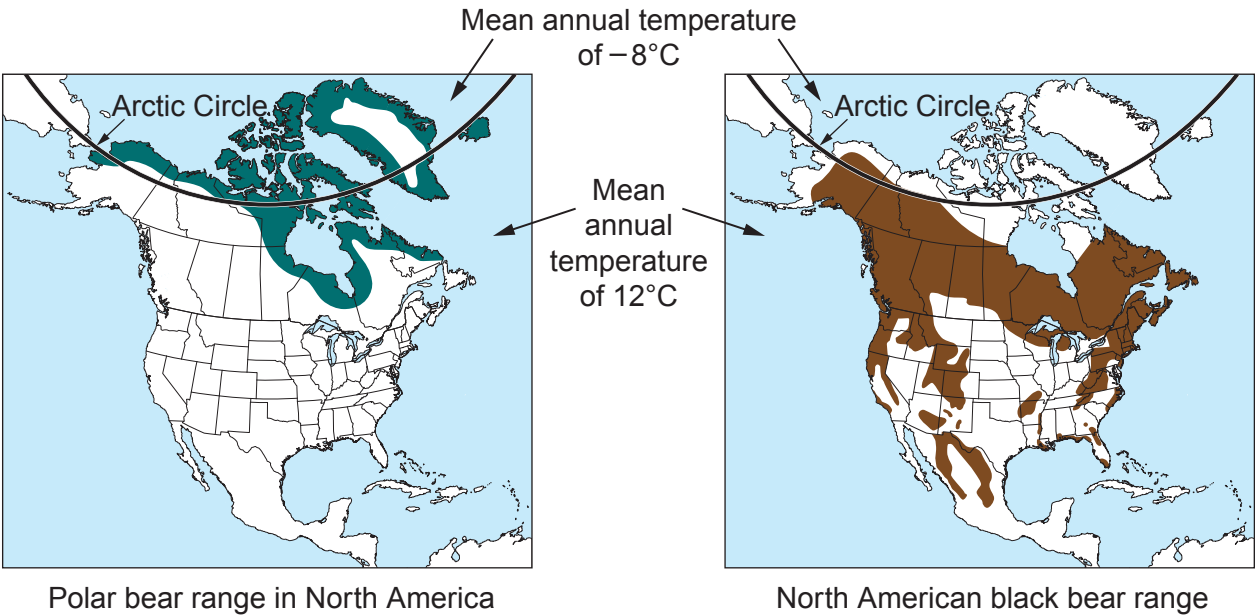
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- (c) Adult male polar bears have a mass of up to 450 kg. The male North American black bear is much smaller with a maximum mass of 158 kg. The shaded areas on the maps in **Image 4.2** show the range of these two bears in North America.

Image 4.2



Use the maps in **Image 4.2**, and your own knowledge, to suggest a reason why the mass of an adult male polar bear is nearly three times as great as that of an adult male North American black bear. [2]

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